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ALICE L WALTON SCHOOL OF MEDICINE

Antivirals 2 of 2: Drugs to Treat Influenza Virus and Respiratory Syncytial Virus Infection

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Adapted from: Drugs to Treat Influenza Virus and Respiratory Syncytial Virus Infection

Listen to this Brick
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Learning Objectives (5) **Versions**

After completing this brick, you will be able to:

1. Describe the mechanism of action and mechanism of resistance of antiviral drugs used in the treatment of Influenza and RSV infections.
2. List the indi eatment of influenza and RSV infections.
3. Discuss the place in therapy for antiviral drugs used in the treatment of RSV infections.
4. Recognize the common adverse reactions associated with influenza and RSV infections.
5. Identify the common drug interactions associated with influenza and RSV infections.

CASE CONNECTION

"I feel awful," reports AP, a 40-year-old man who is at home with classic flu symptoms of fever, chills, myalgia, and cough. Your town is immersed in influenza season; you are asking most patients with flu-like symptoms to speak with you by phone or telemedicine visit rather than come into the office. He has no significant medical history and is taking no medications. His symptoms started about 30 hours ago. He asks, "Would it do any good to take one of those antiviral medicines?"

Would an antiviral do AP any good? Consider your answer as you read, and we'll revisit at the end of the brick.

GO TO CONCLUSION ↓

How do antivirals for influenza and RSV work? (LO#1)

Neuraminidase inhib mainstay of treatment for managing the flu, with oseltamivir being the current drug of choice. Resistance to neuraminidase inhibitors has occurred, but the majority of strains are still susceptible.

Oseltamivir and zanamivir function by blocking neuraminidase, a viral enzyme that releases the budding viruses from an infected cell's outer membrane (Figure 1). This reduces the rate of newly formed viruses traveling to healthy cells and slows the overall progression of infection.

Ribavirin is the only drug that is used for RSV. Ribavirin is a broad-spectrum nucleoside antiviral that is used to treat many other viral infections including hepatitis C. Ribavirin is a guanine nucleotide synthesis inhibitor that functions by mimicking purine structure. It competitively inhibits nucleic acid synthesis of viruses by becoming incorporated into the growing nucleic acid strand and halting elongation. As a guanine analog, it is effective against viral RNA polymerase, which makes it useful in treating RSV and other RNA viruses such as hepatitis C. By inhibiting nucleic acid synthesis, viral replication is decreased.

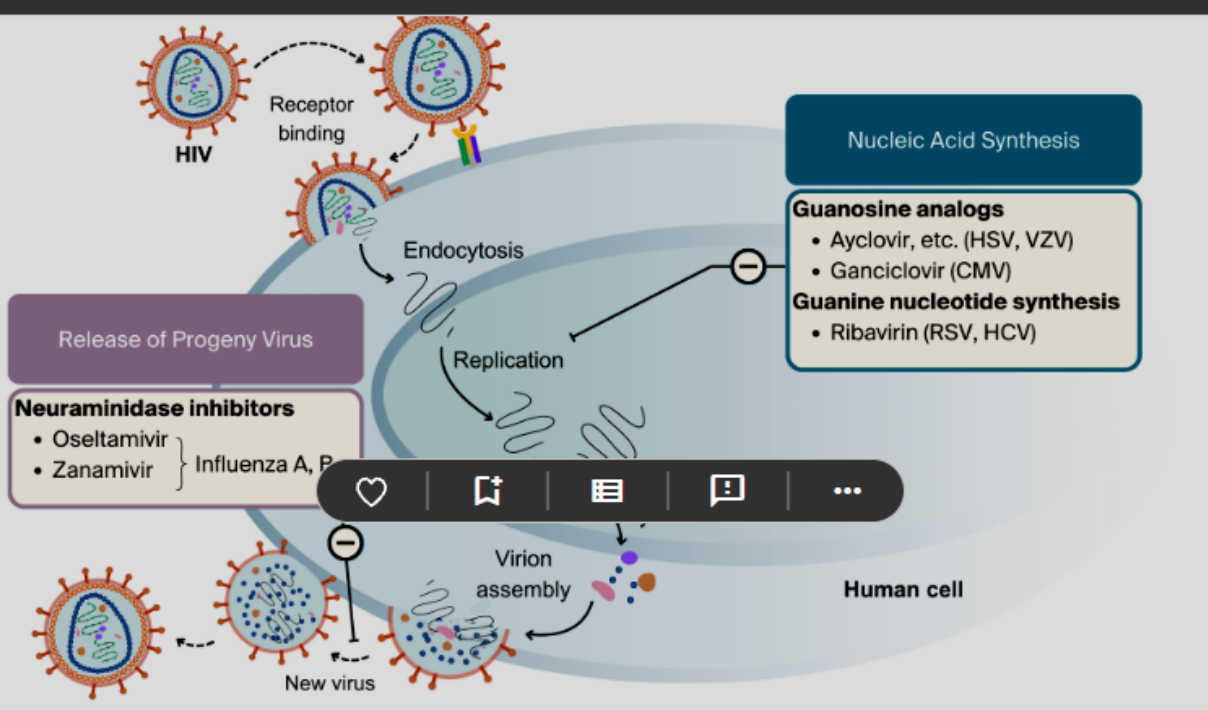


Figure 1

Some influenza strains can demonstrate resistance through various mutations, either with or without exposure to medication. The reduced susceptibility of circulating influenza viruses to neuraminidase inhibitors is mainly due to mutations in the neuraminidase which directly reduce inhibition of enzyme activity.

CLINICAL CORRELATION

As you have learned, viruses are constantly changing and mutating. This can make antiviral drug treatment very challenging. One example, is the H275Y mutation of the influenza virus. This specific genetic change in the neuraminadase gene at amino acid 275 changes a Histadine (H) to Tyrosine (T). This mutation confers resistance to the antiviral drug oseltamavir. This is just one example of a known influenza mutation.

How does ribavirin inhibit viral replication?

When are antivirals indicated? (LO#2)

Influenza and respiratory syncytial virus (RSV) are two common causes of viral respiratory infections that both tend to peak in the winter season. As with other viruses, influenza and RSV rely on host organelles for survival and replication. Supportive care is the foundation of treating these infections because it is most important to keep a patient stable. The health care provider must ensure that the patient maintains hydration, oxygenation, and a controlled temperature. Once these are addressed, antiviral agents can be added. Since influenza and RSV are self-limiting infections, the goal of using antivirals is to minimize symptoms, infectivity, and duration of illness.

Current guidelines recommend antiviral medication for those with *severe influenza requiring hospitalization* or those with influenza and another condition that increases the risk for complications. Some of the risk factors for progressing to severe influenza include, but are not limited to, comorbid diabetes, asthma, or chronic kidney or liver disease; age of 65 or more years; pregnancy or recent delivery (within 2 weeks postpartum); and immunosuppression.

Antiviral medication can also be given to relatively healthy patients at the discretion of the clinician. ***These drugs have maximum effectiveness when started within 48 hours of symptom onset. After this window, the drug does not provide any additional benefit.***

CLINICAL CORRELATION

Why does oseltamivir have to be given within 48 hours of symptom onset?

Remember the mechanism of action above it blocks neuraminidase from budding viruses from an infected cell's outer membrane. Reducing the rate of newly formed viruses traveling to healthy cells. After the 48 hour mark the rate of newly formed viruses is very low, to the point that a neuraminidase inhibitor like oseltamivir would, essentially, be ineffective because the virus is no longer budding. This is an important point to keep in mind for patients who test positive for the flu.

Most often, RSV is self-limiting, however, some patients may benefit from antiviral therapy with ribavirin. This will be discussed further in the next section.

When is an influenza?

What Are the Current Treatment Guidelines for Respiratory Syncytial Virus Infection? (LO#3)

Currently, the preferred treatment of RSV infection is no treatment, since it is a self-limiting infection. Management consists of supportive care to maintain oxygenation, replenish fluids, and manage symptoms. In those with lower respiratory tract infection who present with cough and wheezing, bronchodilators and steroids may result in symptom relief, particularly if the patient has underlying reactive airway disease (i.e., asthma or COPD).

Antiviral therapy with ribavirin is only indicated in patients who are severely immunocompromised, such as transplant patients or those with certain types of cancers. Given the risks of ribavirin therapy (see next section), antiviral therapy for these patients should be carefully considered and individualized.

CLINICAL CORRELATION

As you have learned, RSV is most severe in the very young (i.e., newborns) and elderly populations (i.e., >60 years). Most often, as mentioned above, RSV is self-limiting and does not require medication therapy outside of supportive care.

In May 2023 the FD > 60 years old (with risk factors for severe disease) and pregnant women. This has significantly reduced the rate of RSV infections in both of these vulnerable populations.

When is ribavirin therapy indicated for RSV infection?

What Are the Adverse Effects of these medications? (LO#4)

Antiviral medications are generally well tolerated. Neuraminidase inhibitors typically have adverse effects related to their route of administration. Oseltamivir is given orally and mainly causes headache and gastrointestinal upset. Zanamivir is an inhaled aerosol powder and primarily causes airway irritation.

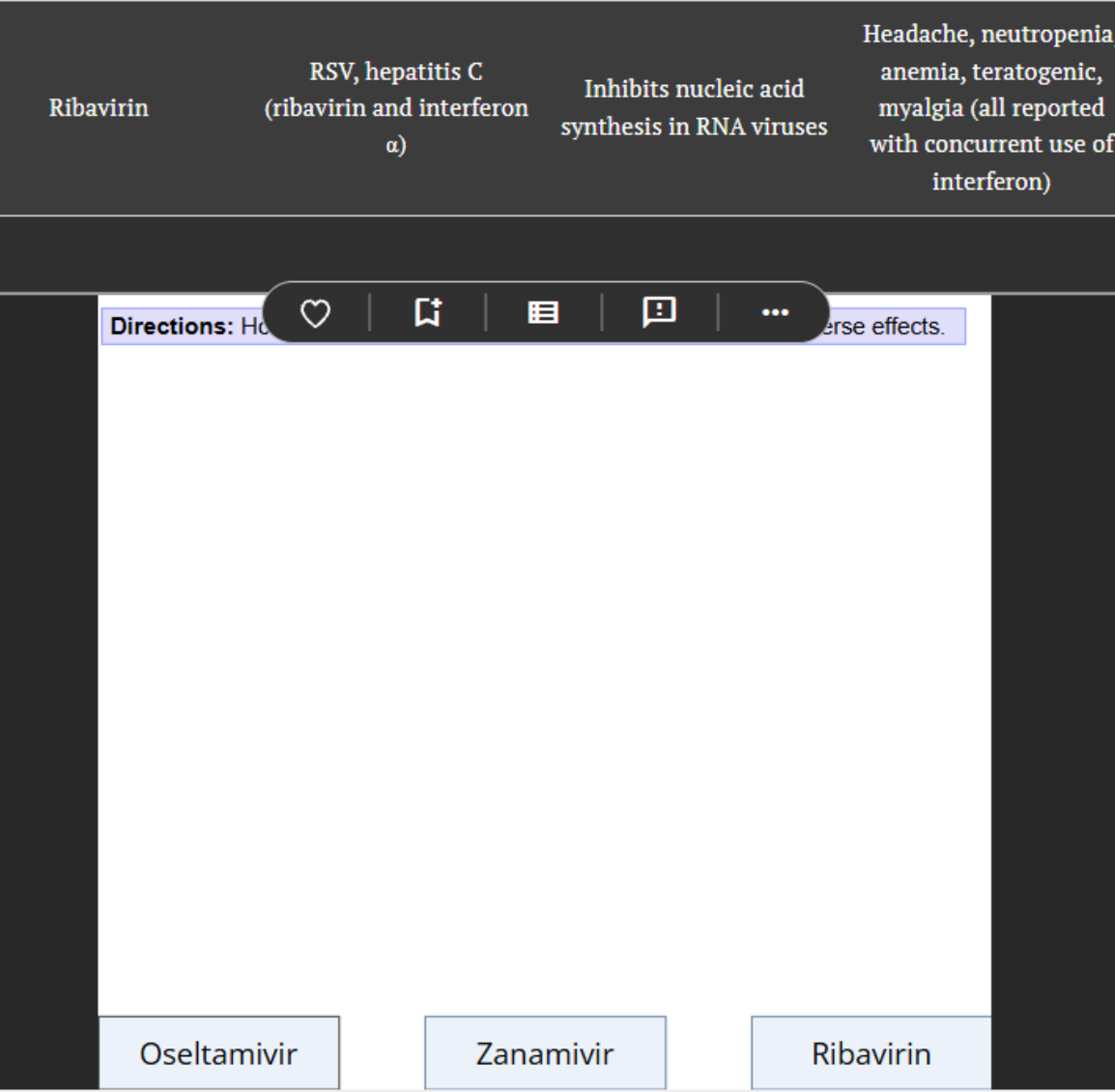
Ribavirin is associated with anemia, hemolytic anemia, and neutropenia. The hemolytic anemia is a black box warning issued by the FDA. It is believed that anemia resulting fro sting cardiac disease and trigger myocardial infarction. Ribavirin must be used with caution in patients with heart disease.

The US Food and Drug Administration has classified ribavirin as a category X drug. This means it is contraindicated for use in pregnant patients because it is a proven teratogen (a substance that causes birth defects when a pregnant mother is exposed to it), and any benefit it provides is outweighed by its risks.

Table 1 provides a summary of the medications discussed in this brick.

Table 1

Drug/Class	Treatment	Mechanism of Action	Adverse Effects
Neuraminidase inhibitors			
Oseltamivir	Influenza	Inhibits viral neuraminidase	Headache, nausea, vomiting
Zanamivir	Influenza	Inhibits viral neuraminidase	Headache, sore throat, myalgia, nasal/airway irritation
Guanosine nucleotide analog			
Ribavirin	RSV, hepatitis C (ribavirin and interferon α)	Inhibits nucleic acid synthesis in RNA viruses	Headache, neutropenia, anemia, teratogenic, myalgia (all reported with concurrent use of interferon)


Directions: Ho erse effects.

What is a major adverse effect of ribavirin?

What are the drug interactions associated with these medications? (LO#5)

Antiviral Agents may diminish the therapeutic effect of the influenza virus vaccine. Avoid administration of live influenza virus vaccine within 2 weeks before or 48 hours after administration of antiviral agents.

Ribavirin may diminish the anticoagulant effect of Vitamin K Antagonists (i.e., warfarin). If a patient requires therapy with ribaviri they will require increased doses or alternate antic complete their course of antiviral.

vitamin K

An antihemorrhagic factor whose activity is associated with compounds derived from naphthoquinone.

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- Malabsorption Syndromes

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CASE CONNECTION

Thinking back to AP, would an antiviral do AV any good

It seems very likely that AP has influenza. You say, "Th treat influenza. Their benefit is not quite so obvious co conditions, like regular antibiotics for a regular infectio work quickly, and the effect can be obvious. When you take a flu antiviral, the most you can expect to gain is feeling better a day sooner than if you were not taking medicine. Of course, when you feel awful, feeling better a day sooner can sound pretty good! The drugs work best if started early. By the time we would get a dose in you, you may already be past the window of opportunity to make a difference. And like most drugs, they do have some side effects, although they are not usually a serious problem. I have to say, I'm not so sure in your case they will have much benefit." You and AP discuss the pros and cons, and he decides that he would like you to call in a presc

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Summary

- Neuraminidase inhibitors (oseltamivir and zan choice for influenza treatment. They reduce se duration, and viral shedding, but are only indic symptom onset.
- Neuraminidase inhibitors should be given to pa symptoms requiring hospitalization and patien put them at risk for developing severe influenz
- Neuraminidase inhibitors are usually well tolerated but may cause headache, myalgia, and gastrointestinal upset.
- Ribavirin, a guanosine nucleotide analog, has a broad spectrum of antiviral activity. It acts against viral RNA polymerase and can be used in severe RSV cases.
- RSV is, most often, self-limiting and primarily managed with supportive care.
- Ribavirin can cause hemolytic anemia, which may exacerbate preexisting heart conditions. Ribavirin is teratogenic and contraindicated in pregnancy.

Review Questions

1. A 16-month-old infant presents to the emergency department with fever, cough, and dyspnea. He was found to be hypoxic and placed on supplemental oxy Which of the following medications is appropriate for treatin

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A. ○ Amantadine
B. ○ Oseltamivir
C. ● Ribavirin
D. ○ Rimantadine
E. ○ Zanamivir

Hide Explanation

The correct answer is ribavirin (C). Ribavirin may be used in severe cases of RSV. Ribavirin has activity against many RNA viruses including RSV due to its activity against viral RNA polymerase. Ribavirin acts as a guanine analog and competes with nucleotides for incorporation into a growing RNA strand. Amantadine (A) is an M2 ion channel blocker. It was formerly used to treat influenza but has been discontinued due to high rates of viral resistance. Oseltamivir (B) is a viral neuraminidase inhibitor used in the treatment of influenza. Rimantadine (D) is an M2 ion channel blocker like amantadine. Zanamivir (E) is a viral neuraminidase inhibitor like oseltamivir.

2. A 42-year-old man comes to the clinic with 3 days of fever, myalgia, and cough. Influenza testing in the office is positive, and the patient begins treatment. Which erse effect of treatment?

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A. ○ Hemolytic anemia
B. ○ Myocardial infarction
C. ○ Nasal irritation
D. ● Nausea
E. ○ Sore throat

Hide Explanation

The answer is nausea (D). The drug of choice for treating the flu is oseltamivir, which is administered orally and thus can cause gastrointestinal disturbances. Nasal irritation (C) and sore throat (E) can be seen with the inhalation of zanamivir. Ribavirin can cause hemolytic anemia (A), and it can worsen the effects of heart disease (including myocardial infarction [B]), thus it is contraindicated in patients with heart disease.

3. What is the primary action of neuraminidase inhibitors?

A. ● blocking viral enzyme
B. ○ reducing viral replication
C. ○ enhancing
D. ○ inhibiting viral entry
E. ○ promoting cell recovery

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Hide Explanation

The correct answer is blocking viral enzyme (A). Neura by blocking neuraminidase, a viral enzyme that release infected cell's outer membrane. Reducing viral replicati action, not the primary action itself. Enhancing immune direct action of neuraminidase inhibitors. Inhibiting viral entry (D) is not the mechanism of action for neuraminidase inhibitors. Promoting cell recovery (E) is not the primary action of these drugs.

4. What should be avoided within 2 weeks before or 48 hours after antiviral agents?

A. ● Live influenza virus vaccine
B. ○ Ribavirin
C. ○ Warfarin
D. ○ Vitamin K

Hide Explanation

The correct answer is live influenza virus vaccine (A). A the therapeutic effect of the influenza virus vaccine, so this timeframe. Ribavirin (B) is an antiviral agent itself, an anticoagulant and not related to the timing of antivir related to the administration of antiviral agents.

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5. When is antiviral medication most effective for i

A. ● Within 48 hours of symptom onset
B. ○ After 48 hours of symptom onset
C. ○ At any time after symptom onset
D. ○ It is not effective

Hide Explanation

The correct answer is within 48 hours of symptom onset (A). Antiviral medications have maximum effectiveness when started within this time frame. After 48 hours (B), the drug does not provide additional benefit. At any time (C) is incorrect as timing is crucial for effectiveness. It is not effective (D) is incorrect because antivirals for influenza can reduce the disease burden if they are given within 48 hours.

How would you rate this

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Review Learning Obje

Check off the objectives when you feel comforta

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02 ☑ ~~List the indications for antiviral drugs used in the treatment of influenza and RSV infections.~~
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Objective Confidence: 5/5